Gallo Sai Siddartha

Mobile Computing CIA-1 192425093

3/12/26

1A) Given:-

Area of metropolitan region = 1200 km^2

cell Radius (R) = 1.8 km

Available Spectrum = 48 MHz

Full Duplex Channel = 200 kHz

Cluster Size (cell 1) = N_1 = 7

Cluster Size (cell 2) = N_2 = 4

Reduction percentage = 15% = 0.15

Minimum Reuse Ratio (θ) = 4.5

Step-1:- Diameter of the cell

$$A_{cell} = \frac{3\sqrt{3}}{2} R^2$$

$$= \frac{3\sqrt{3}}{2} (1.8)^2 = 2.598 \ (3.24)$$

$$A_{cell} = 8.4175 \ km^2$$

No. of cells:-

$$N_c = \frac{1200}{8.4175}$$

$$N_c = 149.175$$

Total Available channels

$$C = \frac{\text{Total spectrum}}{\text{Band width}} = \frac{48 \times 1000}{200}$$

C = 240 channels

After 15% reduction

$C_{effective} = 240 \times 0.85$

$C_c = 204$ channels

Theoritical & Effective Channels

N = 7

$C_{Th} = \frac{240}{7} = 24.29$

$C_{eff} = \frac{204}{7} = 29.19$

N = 4

$C_{Th} = \frac{240}{4} = 80$

Effective channels

$C_{eff} = \frac{204}{4} = 51$

Channel reuse distribution

$D = R\sqrt{3N}$

if N = 7

$D = 1.8\sqrt{3 \times 7}$

$= 1.8\sqrt{21}$

$= 8.25$ km

if N = 4

$D = 1.8\sqrt{3 \times 4}$

$= 1.8\sqrt{12}$

6.24 km

2A) Given

No. of cell boundaries = 15

speed = 90 km/h

NSS path Switching = 80 ms

BSS handoff = 120 ms

authentication delay = 350 ms

Roaming Registration = 250 ms

Recovery delay for each handoff = 400 ms.

Total Delay = NSS Delay + BSS handoff Delay

= 80 + 120

= 200 ms

Converting into seconds

Total Delay = $\frac{200}{100}$ = 2 sec.

Total Recovery Time

= Recovery delay for one handoff × No. of handoff

= 400 × 3

= 1200 ms

Converting into seconds

= 1.2 sec

Total Delay of Registration & Authentication

= 350 + 250

= 600 ms

24) (i) Given

No. of Base Transceiver Stations (BTS) = 12

No. of sectors = 3

No. of Transceivers per sector = 4

No. of timeslots per sector = 2

No. of subscribers per sector = 1200

Average call holding time = 90 seconds

Abis interface of each channel = 16 kbps

single E1 link = 2.048 Mbps

Total No. of Transceivers

$= 4 \times 3 = 12$

Total No. of Timeslots

$= 2 \times 3 = 6$

Total No. of Subscribers $= 1200 \times 3 = 3600$

(ii) Given

spectrum = 10 MHz

FDMA → channel frequency = 30 kHz

N = 7

TDMA → channel frequency = 200 kHz

Users = 8

N = 4

CDMA → Channel frequency = 1.25 MHz

chip rate = 1.2288 Mcps

Voice Activity factor = 0.4

frequency Reuse Efficiency = 0.6

Universal frequency reuse = N = 1

E_b/N_0 = 8

Voice Data Rate = 9.6 kbps.

Channel :-

$$C = \frac{\text{Total spectrum}}{\text{Channel frequency}} = \frac{1000}{30} = 333.33$$

Channel per cell

$$C_{cell} = \frac{333}{7} =$$

TDMA Total Channel

$$\text{No. of Channel} = C = \frac{10000}{2000} = 50$$

Total users = 50 × 8

= 400 users.

channel per cell = $\frac{400}{4}$ = 100 channels.

3A) ~~Given:~~

P% Gain = $\frac{\text{Chip Rate}}{\text{Data Rate}}$

$= \frac{1228}{9.600} = 1$

CDMA Model Efficiency : $k = \frac{PG}{Eb/No} \times V \times F$

$= \frac{128}{8} \times 0.4 \times 0.6$

$= 10 = 3.84$

Total CDMA : $C = \frac{10}{1.25} = 8$

Users $= 8 \times 3.84$

FDMA $= \frac{30.72 - 47.57}{47.54} \times 100 = -35.64$

TDMA $= \frac{30.72 - 100}{100} \times 100 = -69.28$

Compared to TDMA, FDMA use better results

4A) Given:-

No. of files = 240

each file size = 18 MB

Theoritical data rate = 20kbps

Reduced capacity = 18%

Effective throughput = 85%

Packet Retransmission rate = 12%

No. of interruptions = 8

Time for each interruption = 2.5 seconds.

Total Data = No. of file x data of each file

= 240 x 18

= 4320

Data for 8 interpolation R = 4320 x 8

Total theoritical Data rate

= 20 kbps x 4

= 80 kbps

Data rate with 18% percent reduced capacity

R_{red} = 80 x 0.82

= 65.6

Data rate with 85% effective throughput

R_{eff} = 80 x 0.85

Re = 68

Data Volume =

D_1 = 4320 x 1.12

= ~~38~~ 4838.4

Mobility =

T = 8 x 2.5

= 20

Tranda time =

$$\text{Time} = \frac{\text{Data Volume}}{\text{throughput}}$$

$$= \frac{4838.4}{68}$$

Time = 71.153 sec

54) Given:-

Size of software update = 120 GB

No of devices = 2000

Network Bandwith per connection = 9.6 kbps

GPRS effective throughput = 64 kbps

retransmission traffic = 10%

5G Network Bandwith = 25 Mbps

% of protocol overhead = 4%

Data volume per device = $120 \times 1000 = 120000$

$= \frac{120000}{2000} = 60$ MB

GSM transfer time per device

$\frac{480 \text{ Mb} \times 1000}{9.6 \text{ kbps}} = 50,000$ s

$= 13.89$ hours

GPRS transfer time

$= \frac{480 \times 1.10 \times 1000}{64} = 8250$ s

$= 2.29$ hours

effective 5G throughput $= 25 \times (1 - 0.04)$

$= 24$ Mbps